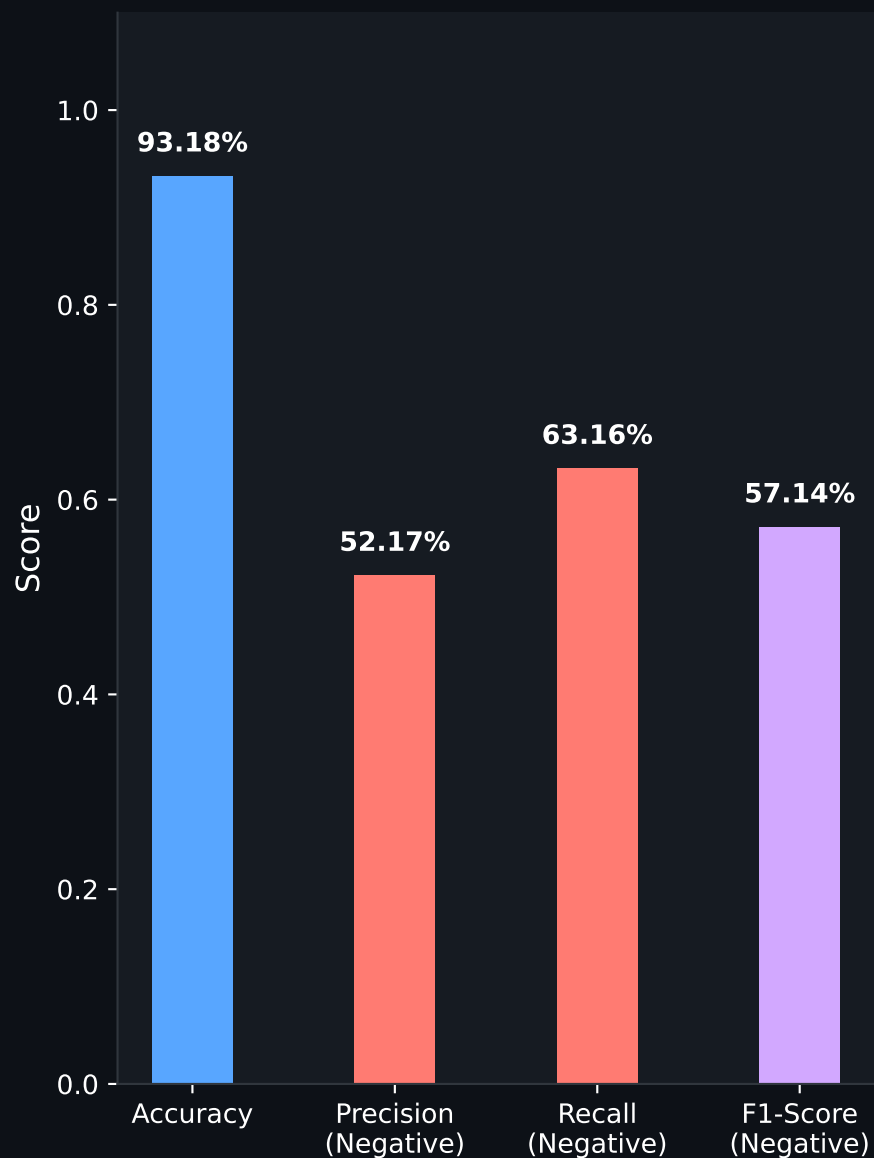
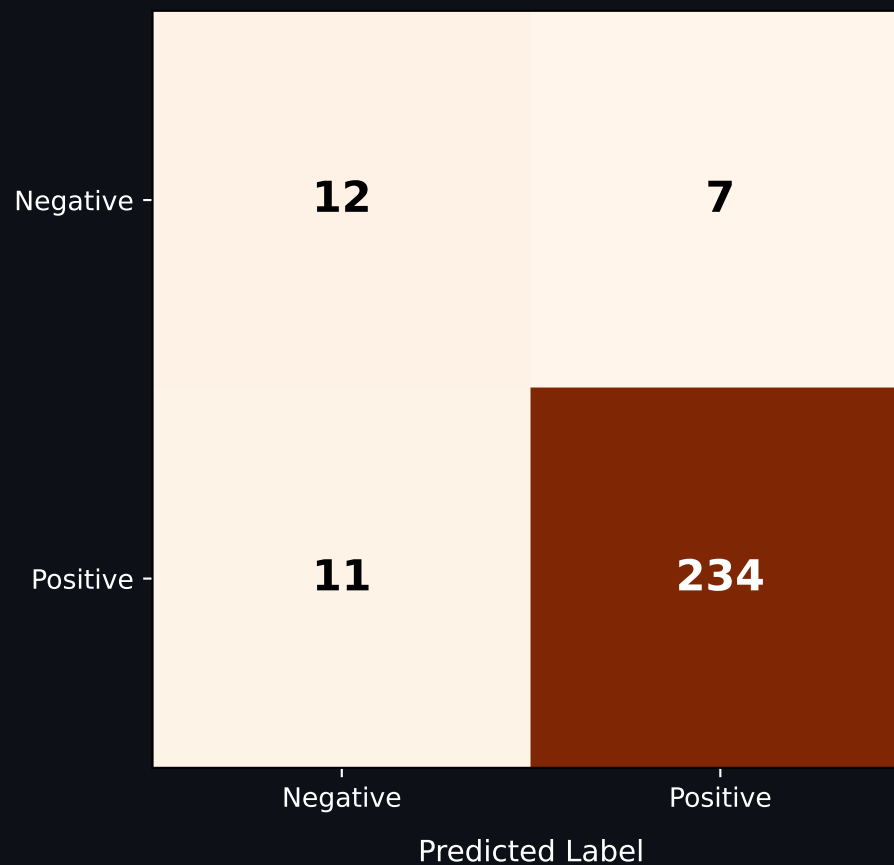


Amazon Sentiment Classifier: Project Overview & Metrics

**Performance Metrics
(Focused on Negative Class)**



Confusion Matrix



Honest Metrics: Custom Dataset Failure Analysis

Word Feature Coefficients
(Red: Negative | Blue: Positive)



Why Accuracy is a Misleading Metric Here:

- Positive reviews make up 92.8% of the dataset.
- A model predicting 'Positive' for everything achieves 92.8% accuracy but fails to detect a single negative review (Recall = 0.0%).
- Our model achieves 93.2% accuracy, but crucially catches 63.2% of the Negative class (Recall) with 52.2% Precision.

- NLP Model Unreliability Case Studies:
1. Double Negations (e.g. 'not disappointed', 'no scratches')
 - Model fails by reading negative components separate from modifiers.
 2. Sarcasm (e.g. 'stand is wonky... better off using peanut butter')
 - Model focuses on words like 'better', completely missing irony.
 3. Multi-Clause mixed comments (e.g. 'product is awesome but slow app')
 - Single word tokenization fails to model clause hierarchy.
 4. Rating-Text Mismatch (e.g. review text is 'Good' but rated 2.0)
 - Model is correct textually, but marked wrong due to bad label.